ELM: full Xe–He production
4 mm pitch, 2.5 mm holes

NAUSICAA

Quantity	Value
Events	8000
Mean detected optical PE/event	505,999.6
SiPM optical charge loss (%)	0.06681
ASIC charge loss (%)	0.00975
Events with clipping (%)	2.288
Input pulses with clipping (%)	0.00099
Retained events	7817
Retained fraction (%)	97.713
Largest optical trigger count/bucket	8,842
Largest optical SiPM charge/bucket (PE)	8,671.49
Maximum expected distinct-cell occupancy (%)	33.33

Stage	All-event RMS (%)	Retained RMS (%)
Detected optical PE	0.19530	0.19571
After SiPM, optical charge	0.20242	0.20111
Noisy, before clipping	0.29136	0.29171
After ASIC	0.31563	0.29171

Stage	Mean average	Mean median	Max average	Max median
Detected optical PE	74.959	59.201	1,252.237	831.000
SiPM optical charge	74.887	59.174	1,246.251	830.017
After ASIC	75.040	59.336	1,230.197	829.902

Setup

Full available bb0nu sample: 8,000 events, Xe–15% He only. E/P = 2 kV/cm/bar at 15 bar; 4 mm square-grid pitch, 2.5 mm hole and light-guide diameter, 5 mm EL depth, and 0.5 mm radial xenon gap. Electron reference distances are sampled uniformly over 10–150 cm with deterministic event

seeds. Ion distances are complementary within 150 cm. Source hits and primary-pair realizations are replayed; event reference distance determines diffusion and attachment. No ion or pure-Xe response was generated.

Electron collection is unity. Uniform axial EL with absorbing walls gives analytical hole-exit acceptance. TPB conversion 0.65, forward fraction 0.5 and meridional guide capture $1-1/1.5$ apply once; emitted-VUV to saved-blue probability is 0.01187461. Propagation and NOA 87 coupling are ideal. Saved blue photons are reused by the response stage.

S14160-3015PS: 3×3 mm^2, 39,984 physical cells, 21,808 uniformly illuminated cells under the centered 2.5 mm guide. Packaged PDE is 0.32 (peak-value approximation), gain 360,000 at +4 V and 25 C. Total primary dark rate is 700 kcps over the full device, split by cell count between illuminated and dark-only populations. Statistical recovery uses a provisional 12.5 ns exponential constant and 128 substeps per 250 ns bucket. Recurrent firing is allowed; PDE remains constant with recovery. Crosstalk, afterpulsing and gain spread are deferred.

Definitions

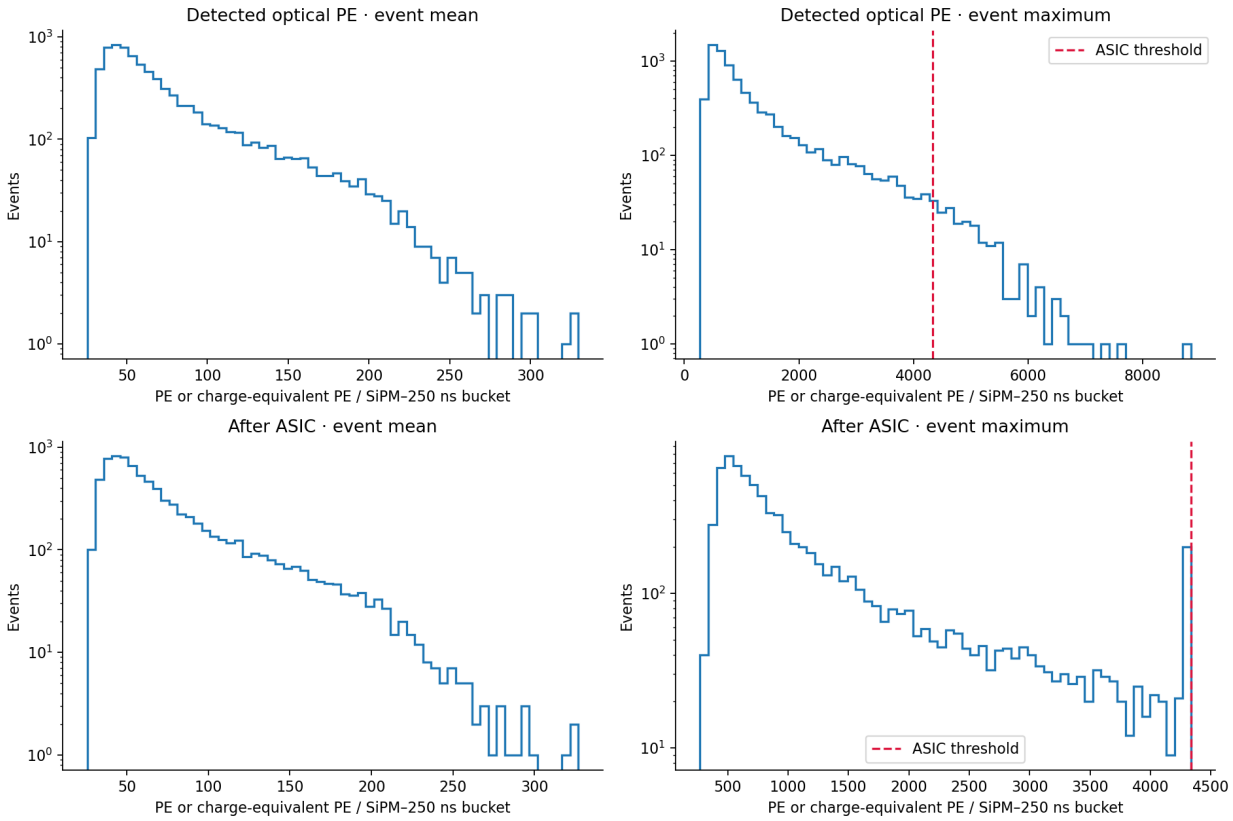
Optical trigger counts are PDE successes. Charge-equivalent PE is charge divided by e $\times$ gain. Distinct optical occupancy is the conditional expected number of cells reached by triggers, divided by the illuminated count. Every event contributes one mean and maximum over its photon-positive input cells, with those same cells used at all stages. SiPM loss compares optical charge with trigger count; ASIC loss compares clipped charge with incoming sensor charge. Electronics adds 3.5417 PE RMS Gaussian noise and clips the bucket-average current at 1 mA (4334.38 PE). Signed outputs are retained.

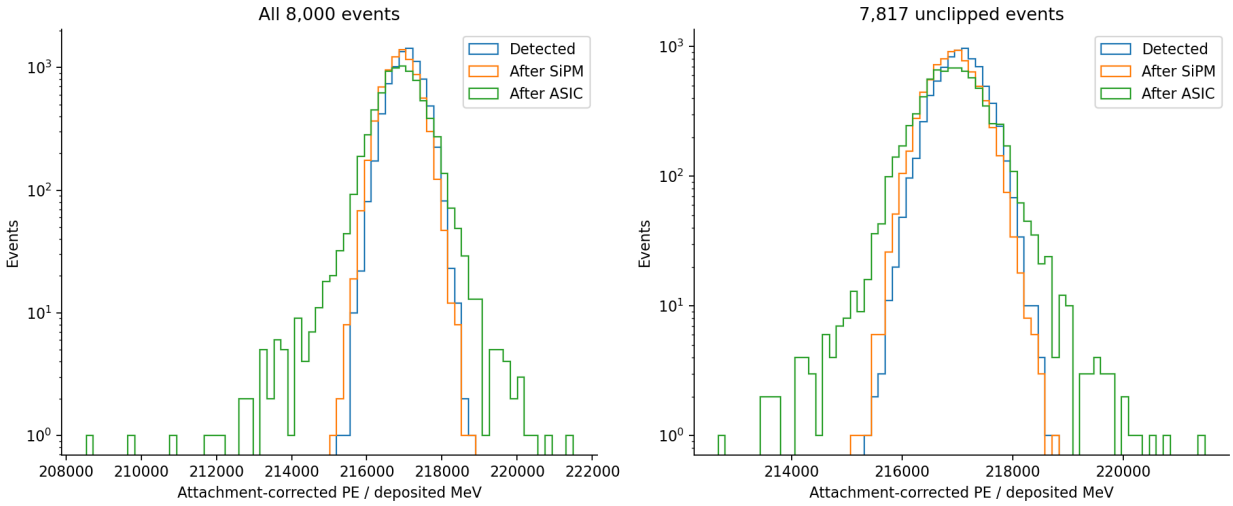
Event sums use all acquired buckets on illuminated channels, including four quiet buckets at each end. Energy diagnostics correct known attachment survival and normalize by deposited MC energy. Optical and sensor widths use optical quantities; noisy totals subtract the expected dark baseline. Reported widths are full-distribution RMS/mean, including tails. The retained sample rejects an entire event if any acquired ASIC bucket clipped. Its width is conditional on that rejection and is reported with the retained fraction. No signal-amplitude cut or SiPM compression correction is applied.

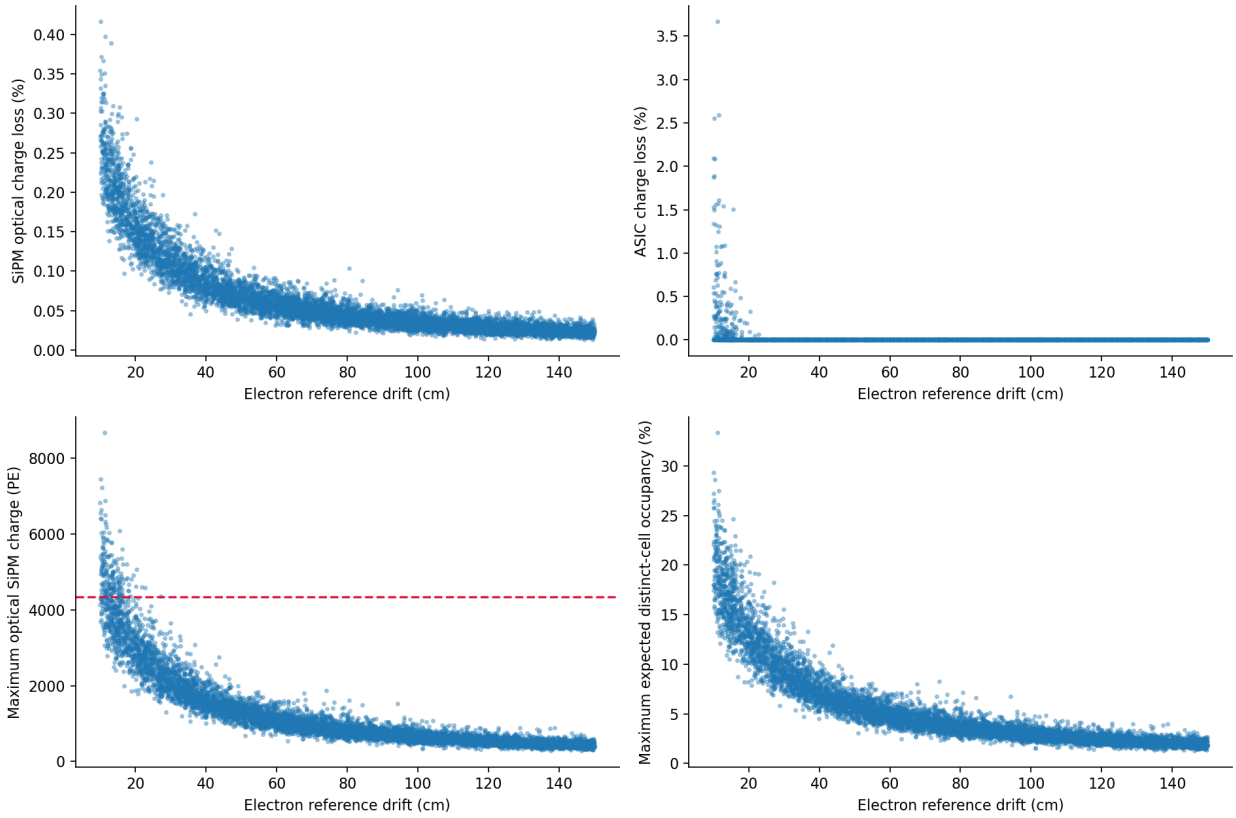
Timing and validation

Photon replay/archive: 70.16 s; four-worker response and assembly: 439.23 s; total timed production: 509.39 s. Independent workers own disjoint 100-event shards and preserve event/channel seeds. Numerical reduction streams quiet-noise moments and retains event summaries; complete bucket histories remain in the independently reusable photon, sensor and ASIC files.

Validation checks source/primary ledgers, selected gas identity, count and charge bounds, PDE sampling, quiet-noise statistics and numerical recovery diagnostics. The streaming multi-shard reduction was checked against the pilot with identical event-level charge and resolution results. Rare feasibility repairs affect the second recovery moment while preserving charge and the first moment; their next-single-trigger variance effect is bounded by the second-moment correction divided by cell count. Its square root is checked against 1% of the electronics noise RMS, and the repair frequency against one per million batches.


Detected optical PE · event mean
Detected optical PE · event maximum
ASIC threshold
After ASIC · event mean
After ASIC · event maximum
ASIC threshold
Events
PE or charge-equivalent PE / SiPM–250 ns bucket


All 8,000 events
Detected
After SiPM
After ASIC
Events
Attachment-corrected PE / deposited MeV
7,817 unclipped events
Detected
After SiPM
After ASIC
Events
Attachment-corrected PE / deposited MeV


SiPM optical charge loss (%)
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
20
40
60
80
100
120
140
Electron reference drift (cm)
ASIC charge loss (%)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
20
40
60
80
100
120
140
Electron reference drift (cm)
Maximum optical SiPM charge (PE)
0
2000
4000
6000
8000
20
40
60
80
100
120
140
Electron reference drift (cm)
Maximum expected distinct-cell occupancy (%)
0
5
10
15
20
25
30
20
40
60
80
100
120
140
Electron reference drift (cm)

Largest event losses

Event ID	Drift (cm)	Max triggers	Max charge PE	SiPM loss (%)	ASIC loss (%)
2004	11.17	8,842	8,671.5	0.3506	3.6657
642	11.50	7,003	6,888.8	0.3972	2.5923
393	10.30	6,731	6,639.6	0.3720	2.5523
5367	10.04	7,566	7,448.3	0.3434	2.0941
5929	10.46	7,349	7,228.4	0.3030	2.0826
878	10.02	6,936	6,832.6	0.3545	1.8753
2968	10.27	6,503	6,397.8	0.3197	1.8904
2932	10.13	6,653	6,556.2	0.4162	1.5264
3170	11.63	6,342	6,265.4	0.3133	1.6121
480	10.66	6,005	5,924.0	0.3254	1.5604
5257	11.39	6,603	6,505.4	0.2874	1.5685
294	10.34	6,122	6,041.9	0.3496	1.4989
4701	12.80	5,511	5,443.4	0.2923	1.5410
1689	15.54	6,169	6,087.8	0.3075	1.5039
5120	10.79	5,966	5,888.9	0.3210	1.3201

Validation ledger

Check	Value
Checked event ledgers	8000
PDE estimate	0.3199989
PDE z-score	-0.267
Quiet noise RMS (PE)	3.54172
Quiet noise mean (PE)	0.000135
Second-moment state repairs	183
Maximum repair / smaller population (ppm)	11.73861
Largest next-trigger RMS impact bound (PE)	0.003426
Charge-variance limits	0

Reproduction

`results/elm_sipm_pitch4mm_production/elm_sipm_pitch4mm_production.h5`; file index `results/elm_sipm_pitch4mm_production/files.toml`. Run `make elm-pitch4-production`; regenerate presentation with `make elm-pitch4-production-report`.

Input cards

```
schema = "nausicaa.electron_detector_spec.v1"
```

```
pressure_bar = 15.0
helium_fraction = 0.15
drift_speed_mm_us = 1.0
lifetime_ms = 15.0
hole_pitch_mm = 4.0
hole_diameter_mm = 2.5
hole_depth_mm = 5.0
sipm_shape = "square"
sipm_side_mm = 3.0
bucket_width_us = 0.25
reduced_field_kV_cm_bar = 2.0
j_variance_ratio = 0.030
j_status = "user_specified"
j_reference = "Oliveira Figure 6; corrected approximate user reading, 2026-11-04"

optical_model = "elm_blue_meridional"
electron_collection_efficiency = 1.0

[optics]
photon_band = "blue_tpb"
tpb_conversion_efficiency = 0.65
forward_fraction = 0.5
guide_refractive_index = 1.5
gap_refractive_index = 1.0
guide_length_mm = 100.0
gas_gap_radial_mm = 0.5
propagation_transmission = 1.0
coupling_transmission = 1.0
electrode_transmission = 1.0
coupling_adhesive = "Norland NOA 87"
guide_length_policy = "fixed_grading_field"
grading_field_kV_cm = 4.65
cathode_potential_V = 1200.0
entry_potential_V = 31500.0
exit_potential_V = 46500.0
sensor_plane_potential_V = 0.0
reference = "docs/elm.tex; isotropic axial EL, absorbing walls, meridional isotropic-in-core
↪  capture"

[pure_xe]
d_T_mm_sqrtbar_per_sqrtcm = 3.5
d_L_mm_sqrtbar_per_sqrtcm = 0.9
yield_gain_photons_per_electron_kV = 137.0
yield_threshold_kV_cm_bar = 0.73

[xe_he15]
d_T_mm_sqrtbar_per_sqrtcm = 1.6
d_L_mm_sqrtbar_per_sqrtcm = 0.75
yield_gain_photons_per_electron_kV = 139.0
yield_threshold_kV_cm_bar = 0.80

[placement]
mode = "uniform_electron_reference"
minimum_electron_drift_cm = 10.0
maximum_electron_drift_cm = 150.0
seed = 2026
reference = "Event reference drift; common diffusion and attachment distance for all hits, original
↪  hit coordinates retained"

schema = "nausicaa.sipm_spec.v1"
response_model = "statistical"
statistical_substeps = 128
sparse_trigger_limit = 16
microcell_count = 39984
illuminated_microcell_count = 21808
microcell_pitch_m = 15e-6
```

```
recovery_s = 1.25e-08
arrival_model = "uniform_in_bucket"
illumination = "uniform_disk_subset"
recovery_pde = "constant"
initial_state = "stationary_dark"
charge_assignment = "avalanche_bucket"
input_photon_band = "blue_tpb"
shape = "square"
side_m = 0.003
pde = 0.32
gain = 360000.0
dark_reference_rate_Hz = 700000.0
dark_reference_area_m2 = 9e-6
crosstalk = false
afterpulsing = false
gain_spread = false
seed = 2026

[acquisition]
channels = "illuminated_in_event"
additional_channels = []
quiet_before_buckets = 4
quiet_after_buckets = 4
empty_first_bucket = 0
empty_last_bucket = 7

[provenance]
device = "Hamamatsu S14160-3015PS, 3x3 mm2; nominal +4 V, 25 C"
pde = "0.32 peak PDE at 460 nm used as provisional TPB-spectrum average; packaged PDE includes
↪  entrance window"
gain = "3.6e5 nominal electrons per primary avalanche; charge per fully recovered avalanche"
dark_rate = "S14160-3015PS nominal primary dark rate 700 kcps at 25 C and +4 V"
window = "S14160-3015PS silicone resin, nominal n=1.57; unit external coupling used in pilot"
datasheet = "papers/SiPM/Hamamatsu_S14160_small_cells.pdf"

recovery = "Provisional 12p5 ns exponential recovery; sensitivity parameter, not a measured 3015PS
↪  recharge constant"
illumination = "Uniform centered 2.5 mm disk on 3x3 mm square: round(39984*pi*1.25^2/9)=21808 lit
↪  cells; remaining 18176 dark-only; full 700 kcps dark rate split by cell count"

baseline_pe = 0.0
bucket_width_s = 2.5e-7
current_limit_A = 0.001
lower_clipping = false
noise = "independent_gaussian_pe"
noise_rms_pe = 3.5416666666666665
schema = "nausicaa.sipm_electronics_spec.v1"
seed = 2027
transfer = "bucket_average_current"

[provenance]
noise = "Same input charge RMS as 0.5 PE at gain 2.55e6; rescaled to gain 3.6e5"
range = "User design input: upper clipping at 1 mA; supersedes the provisional 300 PE limit"
timing = "User 0.25 us buckets; integrated charge divided by bucket duration"
```